

Yunjing "Margaret" Wang

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EDUCATION

- Haverford College** Haverford, PA, United States
• *BS Expected - Physics and Astronomy major; GPA: 3.78* September 2022 - Expected May 2026
Courses: Intro to Astrophysics, Intro to Quantum Mechanics, Modern Theoretical Ideas Of Early Universe Physics (audited this cosmology course,) Waves and Optics, Electronics Waves Optics Lab ,Abstract Algebra I, Linear Algebra, Multivariable Calculus, Plasma Physics, Extragalactic Data Science, Advanced Electromagnetsm, Cosmological Physics, General Relativity, Thermo, Stat Mech, Kinet Theory, Grad level Cosmology and Machine Learning(audited this course,) Topology, Advanced Quantum Mechanics, Obsevational Astronomy, Advanced Classical Mechanics, Condensed Matter Physics(audited this course)

SKILLS SUMMARY

- **Technical:** Python, Mathematica, Linux command-line; TOPCAT; Html.
- **Languages:** English, Mandarin (native), Japanese (limited working capacity)

RESEARCH EXPERIENCE

- Topological Cosmology Research in Copi/Starkman Group** Case Western Reserve University, OH
• *Undergraduate Researcher* Full-time May2025–Aug2025 Part-time Sep2025-present
 - **Research Focus :** Analyzed cosmological topology using wedge/annulus observations of CMB-like fields; compared cylindrical vs. spherical eigenmode behavior under partial-aperture sampling.
 - **Analytical Methods:** Derived eigenmodes with Dirichlet BCs; established orthogonality/normalization and a projection map from full-geometry modes to wedge modes.
 - **Computational Implementation (Python):** Built an NumPy/SciPy pipeline for eigenvalue root-finding, quadrature, and coefficient recovery; produced reproducible plots/notebooks under version control.
 - **Awards & Funding :** Received the KINSC Summer Scholars Fellowship, supporting full-time natural-science research for Summer 2025.
- Commissioning and Calibration of the Xenoscope TPC** University of Zurich, Zurich, Switzerland
• *Student Researcher* Full-time May 2024 - Aug 2024
 - **Xenoscope Calibration:** Assisted in the commissioning and calibration of the Xenoscope Time Projection Chamber (TPC) by analyzing detector performance and optimizing signal readouts.
 - **Data Analysis:** Performed data analysis on calibration runs to identify background noise for potential muon signals.
 - **Collaboration:** Collaborated with a team of physicists to develop testing protocols and contribute to regular progress reports and documentation.
 - **Thinkswiss scholarship:** Awarded the ThinkSwiss Scholarship, selected among thousands of applicants as one of 20 students to receive support for research.
- Independent Astrophysics Research in Galaxy Group** Haverford College, Haverford PA, United States
• *Student Researcher* part-time Sep 2023 - Present
 - **Studied Galaxy Classification:** Conducted research on classifying grand-design and flocculent galaxies to uncover their connection with dark matter.
 - **Machine Learning:** Utilized Python tools to do machine learning on classify galaxy zoo data.
 - **Analyzed Data:** Utilized Python and TOPCAT for comprehensive data exploration and interpretation.
 - **Audited Astronomy Journal Club:** Weekly reviewed scientific papers, gaining insights into the latest research trends.
 - **MNRAS Paper Development:** Monthly Notices of the Royal Astronomical Society Paper in progress.
- Cosmology Research on dark matter relic abundance project** Haverford College, Haverford PA, United States
• *Student Researcher* part-time Dec 2024 - present
 - **Conducted Research:** Simulated and calculated dark matter relic abundance with Python, focusing on varying collision cross-sections.
- Independent Astrophysics Research in Galaxy Group** Haverford College, Haverford PA, United States
• *Student Researcher* full-time May 2023 - Jul 2023
 - **Conducted Research:** Explored spiral structure's correlation with nearby galaxies' baryonic mass fraction.
 - **Achieved 2023 Velay Fellowship:** Women in STEM recognition for Summer Research Grant.
 - **Analyzed Data:** Utilized Python and TOPCAT for comprehensive data exploration and interpretation.
 - **Processed HI Data :** Redacted for Green Bank Telescope.

WORK EXPERIENCE

- **Modern Physics I TA and Grading** Haverford College, Haverford PA, United States
TA and Grader (part-time) Sep 2025 - Present
 - **Graded assignments** : Helped 45 students during TA sessions and evaluated their problem sets weekly.
- **Wave and Optics TA and Grading** Haverford College, Haverford PA, United States
TA and Grader (part-time) Sep 2024 - Dec 2024
 - **Graded assignments** : Helped 45 students during TA sessions and evaluated their problem sets and rewrite weekly.
- **Physics intro Labs Grading** Haverford College, Haverford PA, United States
Grader (part-time) Sep 2023 - May 2024
 - **Graded assignments** : Evaluated 35 student labs weekly.
- **Campus Photography Services** Haverford College, Haverford PA, United States
Photographer (part-time) Sep 2023 - present
 - **Captured campus events**: Effective communication with subjects to capture their vision and convey your ideas.
 - **Edited and processed photos**: Professionally edited and processed images to meet quality and branding standards.

CONFERENCE PRESENTATIONS

- **Research Presentation - October 2023: Title:** "Exploring the Relationship Between Spiral Structure and Baryonic Mass Fraction in Galaxies"
Event: Keck Northeast Astronomy Consortium Student Research Symposium
Description: Engaged both students and esteemed professors and was followed.
- **Poster Presentation - January 2024: Title:** "The Impact of Baryonic Mass Fraction on the Presence of Spiral Structure in Disc Galaxies"
Event: 243rd AAS Meeting
Description: Presented iPoster to participants, delivering clear and concise explanations.
- **Poster Presentation - January 2024: Title:** "The Impact of Baryonic Mass Fraction on the Presence of Spiral Structure in Disc Galaxies"
Event: CUWiP 2024
Description: Presented Poster to participants, delivering clear and concise explanations. Award the best poster.
- **Poster Presentation - October 2024: Title:** "Exploring Dark Matter Detection: The DARWIN Experiment and Xenoscope"
Event: Start Talking Science
Description: Poster in progress on Commissioning and Calibration of the Xenoscope TPC facing to general public.
- **Poster Presentation - March 2025: Title:** "Commissioning of the Xenoscope Time Projection Chamber for the DARWIN Dark Matter Observatory"
Event: APS global physics summit
Description: Poster in progress on Commissioning and Calibration of the Xenoscope TPC

ACTIVITIES AND CLUBS

- **Society of Physics Students** Haverford College, Haverford PA, United States
Co-treasurer Sep 2023 - May 2025
 - **Managed Club Finances:** Oversaw the financial aspects of the club, ensuring support for educational and enjoyable activities for physics students.
- **Society of Physics Students** Haverford College, Haverford PA, United States
Member Sep 2022 - May 2025
 - **Organized Physics-Themed Events:** Planned and executed 4 physics-themed events each semester, securing external funding and fostering an inclusive community within the club.
- **Nerd House(A nerd-dom club for geeks to connect and unite)** Haverford College, Haverford PA, United States
Co-head Sep 2023 - present
 - **Organized Weekly Campus-Wide Events:** Initiated and coordinated weekly inclusive campus-wide events, fostering a strong sense of community and engagement among students.
 - **Established and Improved Internal Regulations:** Actively contributed to the establishment and enhancement of internal rules and regulations within the organization, ensuring smooth and efficient operations.
 - **Facilitated Club Meetings:** Attended weekly club meetings, facilitating communication, collaboration, and decision-making within the group.